SYSTEM OWNER'S HANDBOOK

Current Status and Gaps (Part 8)

Multi-tenant clinic management platform

Status as of 2026-10-28 · code is the source of truth · written for a software-engineer owner

How to read this document

This is part of the **System Owner's Handbook** for a multi-tenant clinic management product, called "the Platform" throughout. It is written for the product owner, a software engineer, and every chapter can be read on its own.

Status labels. Every feature carries exactly one label, based on the code as of 2026-10-28:

- **[Implemented]**: built end to end.
- **[Partial]**: some of it exists; the text says exactly what is missing.
- **[Not implemented]**: planned, seeded, disabled or only mentioned. It is never described as working.
- **[Unverified]**: the code was read but the behaviour could not be confirmed; the text says what is unconfirmed.

Source of truth. The code is the source of truth. Where older documentation disagreed with the code, the handbook follows the code. Nothing was executed while writing: behaviour is described from reading the code.

Figures. Every figure has a number, a caption and a paragraph that describes it in words. Figures captioned "screenshot pending" are placeholders; the paragraph after them describes the screen from the code. The few real screenshots were taken on 2026-10-23 and show a former working name of the product.

Names. "Organization" is one clinic business (the tenant); "location" is one of its branches. File paths and code names appear in a short "Technical references" line at the end of each feature. Example people and places are fictional.

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Chapter 64 — Feature Status Matrix

This chapter lists every feature described in the module chapters (28 to 45), the workflow chapters (46 to 55) and the cross-cutting chapters (56 to 63), one row each, with its status label. It is for the owner who needs to know what is really built before promising anything to a clinic. The status of each row was merged from the notes of the chapters and checked against the chapter text; where a chapter and its notes differed, the chapter wins.

64.1 How to read the matrix

Each row has four columns: the chapter number, the feature as a user would name it, the status label and a one-line note. The four labels mean the following.

- **[Implemented]**: built end to end. The backend and logic exist and a web screen uses them, or it is a backend-only job by design.
- **[Partial]**: some of it exists. The note says what is missing, for example "backend only, no screen calls it".
- **[Not implemented]**: planned, seeded, commented out or only mentioned. It must not be described as working.
- **[Unverified]**: the code could not confirm it. The note says what could not be confirmed.

The workflow chapters (46 to 55) follow one business process across several modules, so many of their rows restate a feature that already has a row in a module chapter. They are listed separately so that each workflow can be judged on its own, and the totals in Section 64.5 report modules, workflows and cross-cutting chapters separately to avoid counting the same feature twice. No application was run while the handbook was written, so every label comes from reading the code.

64.2 Module chapters

64.2.1 Chapter 28 — Patients

Chapter	Feature	Status	One-line note
28	Register a patient	[Implemented]	Two-step dialog, phone unique per org, P-number, optional first appointment
28	Check a phone number before saving	[Partial]	Endpoint and hook exist; no component calls them
28	Browse, search and filter the patient list	[Implemented]	Card + table; UI offers only search, gender, status
28	Open the patient detail page	[Implemented]	Tabs gated by permission and feature; out-of-scope = 404
28	Edit patient details	[Implemented]	Audited diff
28	Activate or deactivate a patient	[Implemented]	status_history row; does not block booking
28	Blacklist a patient	[Partial]	API only (PUT); displayed on Overview; blocks nothing
28	Delete a patient permanently	[Partial]	Fails (500) when billing/lab rows exist (ON DELETE RESTRICT)
28	Import patients from a spreadsheet	[Implemented]	parse → dry run → batches of 500; error report CSV
28	Export the patient list	[Implemented]	10,000-row cap, X-Export-Truncated, audit row
28	Appointments tab	[Implemented]	Appointments + walk-in queue visits

Chapter	Feature	Status	One-line note
28	Visits and clinical history	[Implemented]	GET history; prices stripped without patients.price.view
28	Per-location visit statistics	[Not implemented]	patient_visits never written; endpoint always empty
28	Patient change history	[Implemented]	admin.audit.view; values redacted per permission
28	Dental info	[Implemented]	One row per patient, upsert
28	Vitals	[Implemented]	Append-only, BMI computed
28	Teeth chart (FDI)	[Implemented]	Append-only snapshots; feature off by default
28	Record a treatment from the patient page	[Implemented]	Needs clinical.edit + price.view to see button
28	Patient treatment plans	[Partial]	Backend only (see ch33)
28	Prescriptions tab and legacy quick-Rx	[Implemented]	Legacy patient_prescriptions table still writable
28	Signed consents	[Implemented]	Signature pad, PNG data URL stored in record
28	Documents (Files tab)	[Implemented]	Upload via storage provider, then record
28	Communications log	[Partial]	CrmTab component exists but not mounted
28	Recalls	[Partial]	Backend only; no reminder job
28	Referrals	[Partial]	Backend create/list only; CrmTab unmounted
28	Feedback	[Partial]	Backend create/list only; CrmTab unmounted
28	Insurance	[Partial]	Backend only; feature off by default
28	Billing tab	[Implemented]	Embedded billing module (ch34)
28	Patient portal / CRM feature keys	[Not implemented]	Retired feature keys
28	Patient scope (own/location/org)	[Implemented]	patientScope + PatientScopeGuard

30 rows: 19 Implemented, 9 Partial, 2 Not implemented, 0 Unverified. Patients is the largest module. The core record is built end to end: registration, list, detail tabs, edit, import, export, audit history, dental info, vitals, teeth chart, signed consents and documents. The Partial rows share one pattern: the backend exists but no screen reaches it (phone pre-check, blacklist, recalls, referrals, feedback, insurance, communications log, patient treatment plans). Permanent delete is Partial because it fails with a generic error once billing or lab rows exist. The two Not implemented rows are the per-location visit statistics, which nothing ever writes, and the retired patient portal and CRM feature keys.

64.2.2 Chapter 29 — Appointments

Chapter	Feature	Status	One-line note
29	See bookings and walk-ins in one list	[Implemented]	UNION of appointments and walk-in queue entries, server paging, scoped by appointments.view
29	Browse the calendar by doctor	[Implemented]	FullCalendar premium resource views; drag/resize only scheduled/confirmed; weekly hours shaded
29	See the KPI strip	[Implemented]	Month of selected date; walk-ins and avg wait come from the queue
29	Book an appointment	[Implemented]	Ordered checks plus DB exclusion constraint no_doctor_overlap → DOCTOR_DOUBLE_BOOKED; daily limit default 50

Chapter	Feature	Status	One-line note
29	Check a time slot live and get a suggested time	[Implemented]	POST /appointments/validate, 500 ms debounce, same-day suggestion, up to 50 tries
29	Override booking rules	[Partial]	API flag and appointments.override_rules exist; no screen sends it
29	Reschedule or edit a booking	[Implemented]	Only scheduled/confirmed (APPOINTMENT_LOCKED); emergency switch change on edit is not saved
29	Confirm a booking	[Implemented]	scheduled → confirmed
29	Cancel a booking (with reason codes)	[Partial]	API stores patient_left/other and text; UI sends neither and has no confirm dialog
29	Mark a no-show (by hand or automatically)	[Implemented]	Manual action plus 60 s job using max_late_min
29	Check a patient in (hand-over to the queue)	[Implemented]	QueueOrchestrator.checkInAppointment; detailed in Chapter 30
29	Convert a late booking to a walk-in	[Implemented]	Works; UI rarely reaches it because the no-show job removes the Check in action
29	Respect availability overrides and leave when booking	[Not implemented]	Tables exist in Employees; booking, validation and calendar ignore them
29	Reminders, recurring bookings and online booking	[Not implemented]	Feature keys commented out; notifications are in-app only

14 rows: 10 Implemented, 2 Partial, 2 Not implemented, 0 Unverified. Booking is solid: the list, calendar, live slot check, double-booking constraint and no-show job all work. The gaps are rules the booking engine does not consult (availability overrides and leave), the override-booking flag that no screen sends, and cancel reasons that the API supports but the screen never sends. Reminders, recurring and online booking are not built.

64.2.3 Chapter 30 — Queue

Chapter	Feature	Status	One-line note
30	Open a doctor's queue for the day	[Implemented]	Automatic on arrival or on opening My queue; POST /queue/sessions has no web caller
30	Start, pause and resume the queue	[Implemented]	Pause needs a reason
30	Close the day	[Implemented]	Every remaining patient needs a cancel/transfer decision
30	Switch between smart queue and first come, first served for today	[Implemented]	queue.entry.reprioritize; Admin only by default
30	Doctor-wise tokens	[Implemented]	Counter on the session row, gap-free; prefix from assignment or name
30	Check in a booked patient	[Implemented]	NOT_TODAY, LATE_LIMIT_EXCEEDED, ALREADY_CHECKED_IN guards
30	Register a walk-in	[Implemented]	Quick-creates patient P-xxxxx; ACTIVE_WITH_OTHER_DOCTOR needs force
30	Call the next patient (smart and first-come rules)	[Implemented]	Pure selection engine; 14-scenario spec in both modes

Chapter	Feature	Status	One-line note
30	Select a specific patient	[Implemented]	Engine suggestion recorded; optional required reason
30	Call again (recall)	[Implemented]	Event only, no state change
30	Undo a call	[Implemented]	Back to waiting, same place in line
30	Skip a patient who did not come	[Implemented]	Reception or the session's doctor; also the call-timeout job
30	Rejoin a skipped patient	[Implemented]	end / original_position / manual policy
30	Send for a test and return from the test	[Implemented]	on_hold keeps the token; returns ahead of walk-ins in smart mode
30	Transfer a patient to another doctor	[Implemented]	New token, keeps arrival place; from waiting or in consultation
30	Give urgent priority	[Implemented]	Reason required; no way to remove it
30	Start the consultation	[Implemented]	Creates the patient_encounters row and opens the workspace
30	Complete the visit (and bill it automatically)	[Implemented]	ChargeSourcesService.syncForEncounter in the same transaction (needs billing.invoices)
30	Stop a consultation (abort)	[Implemented]	Appointment cancelled with code other
30	Remove a patient from the queue (cancel)	[Implemented]	Appointment cancelled with code patient_left
30	Force-complete a visit for the doctor	[Implemented]	queue.doctor.any; FORCE_COMPLETED event
30	Print a token ticket	[Implemented]	Pop-up print with a people-ahead estimate
30	Reception board, doctor screen and TV display (reads)	[Implemented]	TV public, no patient names; GET /queue/me unused by the web app
30	Live updates on every screen	[Implemented]	Outbox, relay, Redis, Socket.IO rooms; 10 s polling fallback
30	Safe retries and stale screens (Idempotency-Key and If-Match)	[Implemented]	Keys kept 24 h; VERSION_CONFLICT on a stale version
30	Background jobs	[Implemented]	No-show 60 s, call timeout 15 s, day-end 5 min, clean-up 03:15 server time
30	Queue settings (organization default and location override)	[Implemented]	Only smart mode is fixed per session; other values apply within about 60 s
30	Patient visit history	[Implemented]	Organization-wide, up to 200 rows, patients.view without scope
30	Next-patient preview	[Partial]	Backend only, no screen
30	Queue history (events log)	[Partial]	Backend only (admin.audit.view), no screen
30	Manage the TV display	[Not implemented]	queue.display.manage defined but never checked

31 rows: 28 Implemented, 2 Partial, 1 Not implemented, 0 Unverified. The queue is the most complete module in the Platform: 28 of 31 rows are Implemented. Check-in, walk-ins, smart or first-come calling, tests, transfers, urgent priority, completion with automatic billing, live updates, safe retries and the background jobs all work. The Partial rows are two backend reads with no screen (next-patient preview and queue history). The one Not implemented row is managing the TV display, whose permission is defined but never checked.

64.2.4 Chapter 31 — Consultation

Chapter	Feature	Status	One-line note
31	Start a consultation (encounter created)	[Implemented]	ensureForQueueEntry inside the start-consultation transaction; idempotent per queue entry
31	Re-open a running consultation	[Implemented]	GET /encounters/by-queue-entry/:id; conduct scope own = session doctor
31	See the patient at a glance	[Implemented]	Vitals, 4 previous visits, 3 signed prescriptions with reuse
31	Allergy and medical alerts	[Partial]	Client-side chips from free-text allergies/medical history and latest vitals flags; no structured allergy list or drug check
31	Write the visit notes (auto-saved)	[Implemented]	PATCH /encounters/:id, 800 ms debounce, 4000-char fields; not locked after completion
31	Record the treatments done today	[Implemented]	POST /encounters/:id/treatments; catalog procedure or typed name; no edit/delete
31	Patient record slide-over	[Implemented]	Reuses patient tabs; amounts hidden
31	Prescription inside the consultation	[Implemented]	Draft on first complete line, autosave 1 s, Sign & print; panel not gated by permission/feature
31	Order lab work from the consultation	[Not implemented]	No lab entry point; lab orders have no encounter link
31	Complete the visit	[Implemented]	Flush saves, PATCH complete; appointment completed; charges in same transaction
31	Automatic charges on completion	[Implemented]	ChargeSourcesService.syncForEncounter; only when billing.invoices enabled; needsPricing fallback
31	Void charge of a removed treatment	[Partial]	handleRemoved exists but unreachable: no treatment delete, sync runs once at completion
31	Keep prices away from doctors	[Implemented]	HideTreatmentPricesInterceptor plus VisibleIf on patients.price.view
31	Encounter and treatment records outside the consultation	[Implemented]	GET /encounters/:id, GET/POST /treatments with patient scope
31	SOAP notes and note templates	[Not implemented]	Only encounter text fields; phrase chips fixed in code

15 rows: 11 Implemented, 2 Partial, 2 Not implemented, 0 Unverified. The consultation workspace covers notes with autosave, treatments, the patient slide-over, prescribing, completion and automatic charges. Allergy alerts are Partial because they are only text chips with no structured list or drug check. Voiding the charge of a removed treatment is Partial because no screen can remove a treatment. Ordering lab work from the consultation and structured SOAP notes are not built.

64.2.5 Chapter 32 — Prescriptions

Chapter	Feature	Status	One-line note
32	Browse and filter prescriptions	[Implemented]	GET /prescriptions and /stats, location + patient scope; "this month" uses server clock
32	Write a prescription and save as draft	[Implemented]	POST /prescriptions; RX-{year}-{5 digits} per org, unique platform-wide
32	Fill from a preset	[Implemented]	Builder replaces list; workspace appends and offers advice
32	Sign (finalize) a prescription	[Implemented]	DRAFT/AMENDED only, needs a medicine; best-effort PDF upload
32	Amend a signed prescription	[Implemented]	PATCH on FINALIZED gives AMENDED, version+1, snapshot in history
32	Add/edit/remove one medicine line	[Partial]	Backend only; screens send the full list
32	Duplicate a prescription	[Partial]	Backend only; stores user id as doctor_id (FK to employees), unverified
32	Change history	[Partial]	Rows written; no viewer; read requests undeclared relation changedByEmployee
32	PDF with Urdu (Nastaliq)	[Implemented]	pdf-lib + fontkit-shaped Noto Nastaliq outlines; single page; drafts printable
32	Print layouts (prescription templates)	[Implemented]	Default per location/org; UI creates org-wide only; orientation/logo/reg no. not printed
32	Doctor signatures	[Implemented]	Keyed by user id; delete and by-doctor read have no screen
32	Medicine directory incl. search and seed	[Implemented]	Soft deactivate; seed adds 10 medicines only when empty
32	Prescription presets (treatment-plan-templates)	[Implemented]	About 40 PresetCategory values; not clinical plan templates
32	Prescriptions by patient / visit / appointment	[Partial]	By-appointment lookup has no screen
32	Legacy quick prescriptions (patient_prescriptions)	[Implemented]	Still readable/writable from patient tab; separate numbering, no link to new table

15 rows: 11 Implemented, 4 Partial, 0 Not implemented, 0 Unverified. Prescriptions are largely built, including the Urdu PDF, presets, print layouts, signatures and the medicine directory. The Partial rows are backend-only operations (single-line edits, duplicate, change history, lookup by appointment). Two of them have likely defects that were read from code but not run: the history read and the duplicate action.

64.2.6 Chapter 33 — Treatment plans

Chapter	Feature	Status	One-line note
33	Browse and filter plan templates	[Implemented]	Card grid, search + category; fees stripped without price.view
33	Create a plan template	[Implemented]	Up to 50 procedures in jsonb; not audited
33	Edit a plan template	[Implemented]	Replaces procedure list
33	Hide or show a plan template	[Implemented]	is_active switch
33	Delete a plan template	[Implemented]	Hard delete; treatments.plan_template_id SET NULL

Chapter	Feature	Status	One-line note
33	Starter templates for a new clinic	[Implemented]	12 defaults seeded when org has zero templates (re-seeds if all deleted)
33	Use a template when recording a treatment	[Implemented]	Prefills name/cost, tags treatment; button needs price.view
33	Create a patient treatment plan	[Partial]	Backend only; web hooks unused
33	Update a plan (status, acceptance, costs)	[Partial]	Backend only; free status changes
33	Add items to a plan	[Partial]	Backend only
33	Track progress of plan items	[Partial]	Backend only; nothing automatic
33	Start a patient plan from a template	[Not implemented]	Only UI text/comment claim it
33	Turn plan items into charges	[Partial]	Backend only; no web caller
33	Plan billing summary	[Partial]	Endpoint + API fn + hook; no component renders it

14 rows: 7 Implemented, 6 Partial, 1 Not implemented, 0 Unverified. Plan templates are fully built. Patient treatment plans are the weak half: creating, updating and tracking a patient plan, turning its items into charges and the plan billing summary all exist only in the backend. Starting a patient plan from a template is not built, although some screen text says it is.

64.2.7 Chapter 34 — Billing

Chapter	Feature	Status	One-line note
34	Keep a service catalog	[Implemented]	Org-level services, audited, never deleted
34	Set location and doctor price overrides	[Implemented]	PATCH override and resolved-price endpoints unused by UI
34	Automatic charges from a completed consultation	[Implemented]	syncForEncounter in queue completion tx; removed treatment → VOID or Review
34	Add a manual charge	[Implemented]	Catalog or Other; price policy enforced
34	Create charges from a treatment plan	[Partial]	Backend only, no web call
34	Edit an unbilled charge (price policy)	[Implemented]	Catalog price needs billing.price.override
34	Void an unbilled charge	[Implemented]	Reason 5–300 chars
34	Manage discount types	[Implemented]	Code field not on screen
34	Discount one / many / custom	[Implemented]	Range via AccessContext.decide; custom needs note
34	Ready-to-bill list and billing desk	[Implemented]	Outstanding omits partly paid invoices
34	Collect a payment	[Implemented]	One tx, lock order, idempotent clientRequestId
34	Choose an allocation strategy	[Implemented]	Oldest first (FIFO) and Manual

Chapter	Feature	Status	One-line note
34	Hold and apply advance	[Implemented]	Advance = amount − allocated − refunded
34	Unapply a payment	[Partial]	Backend only
34	Payment accounts and methods	[Implemented]	Auto Cash default; default/last account cannot be deactivated
34	Number invoices, receipts and refunds	[Implemented]	CODE-INV/REC/RFD-YYYY-NNNNN, gapless
34	Issue an invoice	[Implemented]	At collection/advance apply or manual; due days default 7
34	Exclusive sales tax	[Implemented]	Snapshot of location_settings.tax_rate
34	Browse invoices and payment status	[Implemented]	Stored ISSUED/CANCELLED; computed payment status
34	Edit an issued invoice	[Implemented]	Preview in rolled-back tx; revision conflict check
34	Cancel an invoice	[Implemented]	Only when nothing paid; unapply has no UI
34	Refunds	[Implemented]	Always from advance; invoice refund may carry an edit
34	Patient billing summary and statement	[Implemented]	Ledger-based statement
34	Show dues at check-in	[Implemented]	Setting enforced only in web app
34	Daily collection	[Implemented]	Payments page
34	Cash close	[Implemented]	Advisory lock; immutable; variance flag
34	Printing	[Implemented]	Browser print A4/80 mm; English labels, Urdu typed text
34	Billing settings	[Implemented]	Document code locks after first number
34	Expenses	[Implemented]	Edit API-only; no delete confirm; hard delete
34	Credit packages and patient credits	[Partial]	Backend only; expiry cron 01:00 server date
34	Billing activity / audit log	[Implemented]	User/action filters not on screen
34	Nightly invariant check	[Implemented]	02:30 server time, I1–I10, log only
34	Doctor revenue and financial summary	[Implemented]	Daily summary unused; details in Ch 36
34	Commission snapshot on charges	[Implemented]	Cross-reference to Ch 35

34 rows: 31 Implemented, 3 Partial, 0 Not implemented, 0 Unverified. Billing is the deepest module and is almost entirely Implemented: catalog, automatic and manual charges, discounts, collection, advance, invoices, tax, edits, cancellation, refunds, cash close, printing, expenses and the nightly checks. Three Partial rows are backend-only: unapplying a payment, credit packages and plan-based charges. Because unapply has no screen, an invoice that has any payment cannot be cancelled from the web app.

64.2.8 Chapter 35 — Commissions

Chapter	Feature	Status	One-line note
35	Set up or change a compensation plan	[Implemented]	Versioned per assignment, GiST no-overlap, compensation.manage
35	See the current plan and version history	[Implemented]	Compensation tab in employee detail
35	Edit service-wise overrides (commission rules)	[Implemented]	Replace rules of current version; service > category > default
35	Snapshot the rate on each visit charge	[Implemented]	CONSULTATION and TREATMENT charges only; backend by design
35	Earn commission when patient money is applied	[Implemented]	Cumulative target, NET/GROSS, tax pro-rating, fixed amount; backend by design
35	Earn commission on lab work	[Implemented]	Lab-line commissions, net of lab cost; backend by design
35	Preview the doctor share while collecting payment	[Implemented]	doctorShare gated by earnings.all.view and billing.commissions
35	See every doctor's earnings	[Implemented]	/billing/doctor-earnings; GET /commissions has no screen
35	See my own earnings	[Implemented]	/my-earnings, filtered by signed-in user
35	See doctor revenue	[Implemented]	/billing/doctor-revenue, reports.financial.view, not commission
35	Settle commission through payroll	[Implemented]	lockUnsettled, create, markPaid, release
35	Protect the ledger (append-only trigger, I10)	[Implemented]	Trigger in migration; nightly I10 covers charges only
35	Choose a commission plan when adding an employee	[Implemented]	compensation object on POST /employees
35	Manual commission adjustments	[Not implemented]	MANUAL source type reserved, never written

14 rows: 13 Implemented, 0 Partial, 1 Not implemented, 0 Unverified. Commissions work end to end in the backend, which is by design: rates are snapshotted on charges, commission is earned as patient money is applied, lab lines are included and the ledger is protected by a trigger. Screens exist for plans, earnings and the doctor-share preview. Manual commission adjustments are the only gap: the source type is reserved and never written.

64.2.9 Chapter 36 — Dashboard and reports

Chapter	Feature	Status	One-line note
36	See the home dashboard	[Partial]	Real data; money tiles also need billing.financial_summary; mock dashboard.api.ts and ~10 widgets unused
36	Read the finance overview	[Implemented]	/reports/finance/overview, gated finance.overview.view, 12-month trend, warnings

Chapter	Feature	Status	One-line note
36	Read the profit and loss statement	[Implemented]	Collected vs billed basis; costs by own period; payroll base salary only
36	Read the clinic overview	[Partial]	Default Doctor role (own scope) sees sidebar item but API refuses
36	Browse the report library	[Implemented]	37 reports, filtered by permission and own-scope doctor filter
36	Run a report on screen	[Implemented]	5,000 row cap, scope resolution, rupees
36	Download a report as Excel or PDF	[Implemented]	50,000 rows, PDF max 5,000, audit EXPORT row
36	The 37 reports	[Implemented]	15 financial, 22 clinical; 9 categories
36	Daily financial summary	[Partial]	Backend only; /financial-summary redirects to /dashboard

9 rows: 6 Implemented, 3 Partial, 0 Not implemented, 0 Unverified. The finance overview, profit and loss, the 37-report library and Excel and PDF export are Implemented. The home dashboard is Partial because roughly ten unused widgets still read mock data and the money tiles need an extra feature. The clinic overview is Partial because the default Doctor role sees it in the sidebar but the API refuses that scope. The daily financial summary has no screen.

64.2.10 Chapter 37 — Employees

Chapter	Feature	Status	One-line note
37	Browse and filter the employee list	[Implemented]	Assignments at active location, server paging
37	Add an employee	[Implemented]	Platform-unique phone checked; CNIC only by DB index
37	View an employee	[Implemented]	Scoped to active location
37	Edit an employee	[Implemented]	Duplicate phone/CNIC on edit gives generic 500
37	Activate or deactivate an employee	[Implemented]	Assignment is_active; employees.is_active never written
37	Delete an employee permanently	[Implemented]	Refused if other locations or referenced rows
37	Give an employee a staff login	[Implemented]	Employee form and User Admin; PATCH portal-access has no screen
37	Set or update the salary	[Implemented]	Adds compensation plan version from today
37	Set the weekly schedule	[Implemented]	7 days, slot 15/30/45/60; cross-location check never called
37	Record availability overrides	[Partial]	CRUD works; booking ignores them
37	Manage leave requests	[Partial]	Manager-only; approved leave ignored by booking and payroll
37	See the leave summary	[Partial]	Usage tally, no entitlements
37	List employment roles	[Implemented]	GET /employment-roles, employees.view
37	Assign an existing employee to another location	[Not implemented]	DTOs exist, no endpoint
37	Attendance	[Not implemented]	Only in hr.leave description

15 rows: 10 Implemented, 3 Partial, 2 Not implemented, 0 Unverified. Employee records, schedules, salary, staff logins and deletion are built. Availability overrides, leave requests and the leave summary are Partial because they can be recorded but are never read by booking or payroll. Assigning an existing employee to another location and attendance tracking are not built.

64.2.11 Chapter 38 — Payroll

Chapter	Feature	Status	One-line note
38	Generate payroll for a month	[Implemented]	Base + commission, settlement lock, future months refused
38	Month-end automatic generation	[Implemented]	Cron 0 5 1 * * server time; comment says 00:05
38	Create a payroll record by hand	[Implemented]	No assignment or future-month check; commission 0
38	View payroll for a month or employee	[Implemented]	month/year validation 400
38	Mark a record as paid	[Implemented]	Also pays settlement; records payout only
38	Put a record on hold	[Partial]	Endpoint exists; no button, edit dialog status works
38	Edit a payroll record	[Implemented]	Amount is total; floor at commission
38	Delete a payroll record	[Implemented]	Releases pending settlement; refused if commission paid
38	Export a month as CSV	[Partial]	Endpoint only, no button
38	Payslip PDF	[Partial]	Endpoint only, no button
38	Deductions, tax and bank files	[Not implemented]	Deductions fixed at 0

11 rows: 7 Implemented, 3 Partial, 1 Not implemented, 0 Unverified. Payroll generation, month-end automation, editing, deleting and marking paid all work. Putting a record on hold, the CSV export and the payslip PDF are Partial: the endpoints exist but no button calls them. Deductions, tax and bank files are not built, so net pay always equals gross pay.

64.2.12 Chapter 39 — Inventory

Chapter	Feature	Status	One-line note
39	Browse stock and see stock health	[Partial]	Works; no sidebar entry
39	Add, edit and deactivate an item	[Partial]	Initial stock uses movement path (FK risk)
39	Manage categories	[Partial]	API only; no screen creates; seedDefaults never called
39	Adjust stock and keep the movement history	[Unverified]	performedBy gets user id but FK is to employees
39	Manage suppliers	[Partial]	Works; no sidebar entry
39	Raise and track a purchase order	[Unverified]	createdBy FK risk; poNumber global unique vs per-org sequence
39	Receive goods into stock	[Unverified]	Same FK risk via applyMovement
39	Define consumption templates	[Partial]	CRUD works; nothing uses templates
39	Record materials used on a patient	[Not implemented]	ConsumptionRecorder not placed on any screen
39	Daily low-stock and expiry alerts	[Partial]	08:00 server-time cron only logs

10 rows: 0 Implemented, 6 Partial, 1 Not implemented, 3 Unverified. Inventory has no Implemented row. Stock, suppliers, purchase orders and receiving work through the API and have screens, but the module has no sidebar entry. Three rows are Unverified because the code passes a login id into columns that are foreign keys to employees, and this was never run to see whether the writes fail. Recording materials used on a patient has no screen, and the daily alert job only writes to the log.

64.2.13 Chapter 40 — Dental lab (clinic side)

Chapter	Feature	Status	One-line note
40	Keep the platform lab directory	[Implemented]	Super admin; external and in-house labs; in-house lab auto-creates its active link
40	Manage the staff of a lab	[Implemented]	Super admin adds lab members by account email; deactivate, never delete
40	Manage a lab's services, options and prices	[Implemented]	Super admin only; price windows, void, history; lab cannot edit its own catalog
40	Service categories	[Partial]	Create/update endpoints exist, no screen; six dental categories seeded
40	Add a lab to the clinic's lab list	[Implemented]	Invited link; directory shows active external and own in-house labs
40	Set payment terms and patient data level	[Implemented]	full_name or code_only per link; 0-365 days
40	Activate, suspend and reactivate a lab link	[Implemented]	Reason to suspend; only suspending side reactivates
40	Read the lab price list	[Implemented]	Negotiated else list price; margin; AnyMember reads
40	Get a price quote	[Implemented]	LabPriceResolver; quote default day is Asia/Karachi
40	Map a lab service to a clinic treatment	[Implemented]	One default per treatment; needs lab.catalog.manage
40	Save and reuse order templates	[Implemented]	Private or shared; /use endpoint not called by web
40	Create a lab order draft	[Implemented]	Optional idempotency key; pins form and workflow versions
40	Start from a template or repeat a past order	[Implemented]	from-template and repeat endpoints
40	Submit an order to the lab	[Implemented]	One transaction; Idempotency-Key required; price snapshot; revision 1; magic link
40	List and search lab orders	[Implemented]	Scoped by lab.orders.view own/location/org
40	Open an order and print the work order	[Implemented]	QR /o/:orderId; no prices on work order
40	Cancel, hold, resume or reopen an order	[Implemented]	Workflow driven; cancel from draft/submitted/accepted only
40	Revise a submitted order	[Implemented]	New revision, re-priced on original day
40	Talk to the lab: messages and clarifications	[Implemented]	Clinic replies; only lab asks

Chapter	Feature	Status	One-line note
40	Decide the lab's extra charge requests	[Implemented]	Approved item charge adds to lab cost
40	Attach and download files	[Implemented]	50 MB; 10-minute signed URL; patient_shared not surfaced
40	Send, copy and revoke magic links	[Implemented]	Hash only in DB; 24 h default TTL
40	Update on behalf of a lab that does not answer	[Implemented]	Needs lab.orders.create and lab.links.manage; via proxy
40	Follow production and try-in rounds	[Implemented]	Stages from template; rounds; record try-in; decide
40	Approve, reject or deliver finished work	[Implemented]	Delivery starts 12-month warranty
40	Cancel one item	[Partial]	Endpoint only, no screen
40	Ask for a remake and decide who pays	[Implemented]	Warranty claim auto; charges by responsible party
40	Receive a shipment and report a lost one	[Implemented]	QR token; only lab.orders.view needed
40	Send impressions and models to the lab	[Partial]	Endpoint and unused web API function
40	Notifications about lab work	[Partial]	4 seeded events never emitted; patient/lab-contact channels dev-log only

30 rows: 26 Implemented, 4 Partial, 0 Not implemented, 0 Unverified. The clinic side of lab work is broad and mostly Implemented: directory, links, price lists, orders, revisions, messages, files, magic links, production rounds, delivery, warranties, remakes and shipments. The Partial rows are cancel-one-item, sending impressions to the lab, service categories and lab notifications (four seeded events are never emitted).

64.2.14 Chapter 41 — Lab portal

Chapter	Feature	Status	One-line note
41	Sign in as a lab member	[Implemented]	labMemberships in login response; lab-only users land on /lab-portal
41	Work in one lab or switch between labs	[Implemented]	x-lab-id header; LabPortalGuard
41	See the lab dashboard	[Implemented]	Three counters
41	Read the order inbox	[Implemented]	Drafts hidden; patient ref by data level
41	Open an order	[Implemented]	Lab mapper: no patient price
41	Accept an order with an expected date	[Implemented]	Lab admin; creates stages and vendor bill draft
41	Reject an order	[Implemented]	Order-level reject with reason
41	Ask the clinic a question	[Implemented]	Moves order to clarification_needed

Chapter	Feature	Status	One-line note
41	Change the expected date	[Implemented]	Lab admin; reason; accepted/in progress/partially ready/on hold
41	Write a note to the clinic	[Implemented]	Notes only; no status change
41	Acknowledge a revision	[Implemented]	Idempotent
41	Ask for an extra charge	[Partial]	Backend only; no portal form
41	Upload files to an order	[Partial]	Backend only; no portal control
41	Download files the clinic shared	[Implemented]	lab_shared only; 10-minute URL
41	Run the work queue: start, QC pass, QC fail	[Implemented]	Lab admin and technician
41	Assign a technician	[Implemented]	Lab admin; unassign endpoint has no button
41	Ship a try-in or finished work to the clinic	[Implemented]	SHP number, QR slip
41	Receive what the clinic sent	[Implemented]	Portal always records full qty, condition ok
41	Report a shipment lost	[Implemented]	Open lines become missing
41	Dispute a remake request	[Implemented]	Lab admin; only requested status
41	Accept, suspend or reactivate a clinic	[Implemented]	Lab admin; shared transition code
41	Manage the lab's team	[Implemented]	Lab admin; no self change
41	View and edit the lab profile	[Implemented]	Contact fields only
41	See bills, payments and statements	[Implemented]	Read only; drafts hidden
41	Answer an order from a magic link, without logging in	[Implemented]	Accept, reject, ask; public routes; throttled
41	Open an order from the printed QR code	[Implemented]	/o/:orderId resolves lab or clinic
41	Be told about new work	[Partial]	In-app rows exist; no inbox in portal layout found

27 rows: 24 Implemented, 3 Partial, 0 Not implemented, 0 Unverified. The lab portal covers sign-in, order inbox, accept, reject, clarify, production queue, shipping, team and profile, and no-login magic-link answers. Partial rows are charge requests and file upload (backend only) and the new-work notification (rows exist but the portal has no inbox).

64.2.15 Chapter 42 — Lab finance

Chapter	Feature	Status	One-line note
42	Estimate and invoice lab work	[Implemented]	draft to issued, LINV numbers, frozen patient prices
42	Browse lab invoices and see one	[Implemented]	overdue computed in location timezone
42	Void estimate or unpaid invoice	[Implemented]	needs billing.invoice.void too
42	Collect lab payment	[Implemented]	Idempotency-Key required, FIFO auto-allocation
42	Use patient advance	[Implemented]	also automatic at issue
42	Reverse lab payment	[Implemented]	needs billing.refund.issue; refused after refund
42	Credit note	[Implemented]	releases newest allocations to advance
42	Refund advance	[Implemented]	needs billing.refund.issue
42	Payment reminder	[Partial]	queued to outbox, patient audience off by default, nothing delivered
42	Order and patient billing summaries	[Implemented]	read panels
42	Lab billing settings	[Implemented]	tax bp and due days; needs admin.settings.manage
42	Doctor commission on lab work	[Implemented]	collected basis net of lab cost; skipped when billing.commissions off
42	Receivables journals (GL)	[Partial]	written for all lab money, no screen or report reads them
42	Vendor bill drafts at lab accept	[Implemented]	effect create_vendor_bill_draft plus item listener
42	Review, approve, void vendor bill	[Partial]	PATCH bill and add line exist in API only
42	Pay a lab	[Implemented]	allocations over approved bills, journals
42	Use lab advance on bills	[Partial]	endpoint and api client exist, no UI button
42	Reverse a payment to a lab	[Implemented]	mirrors journals
42	Lab statements and reconciliation	[Implemented]	manual line entry, no upload
42	Payables reports	[Implemented]	aging, monthly cost, case profit
42	Lab portal payables (read)	[Implemented]	drafts excluded

21 rows: 17 Implemented, 4 Partial, 0 Not implemented, 0 Unverified. Lab finance has full estimate-to-invoice, collection, advance, credit note, refund, vendor bill, payment and statement flows. Partial rows are backend-only vendor bill editing and lab advance use, payment reminders that are queued but never delivered, and journals that are written but never read by any screen.

64.2.16 Chapter 43 — Workflow engine (Config Studio)

Chapter	Feature	Status	One-line note
43	Manage form schemas	[Implemented]	super admin only
43	Edit form version and preview	[Implemented]	issues do not block saving
43	Publish and retire form version	[Implemented]	service-enforced immutability
43	Clinics read current forms	[Implemented]	AnyMember endpoints
43	Manage workflow templates	[Implemented]	
43	Edit workflow draft and test run	[Implemented]	60 states, 300 transitions
43	Publish and retire workflow version	[Implemented]	rows materialised at publish
43	Manage stage templates	[Implemented]	not versioned, no delete
43	Order state machine from pinned version	[Implemented]	pinWorkflowVersion, transition referee
43	Compare versions and activity	[Implemented]	activity tab on forms only
43	Specialties and organization enablement	[Partial]	endpoints exist, Studio only lists
43	Dental seed	[Implemented]	insert-if-missing at boot

12 rows: 11 Implemented, 1 Partial, 0 Not implemented, 0 Unverified. Config Studio is a super admin tool for form schemas, workflows and stage templates. Everything is Implemented except specialties and organization enablement, where endpoints exist but the Studio only lists.

64.2.17 Chapter 44 — Notifications

Chapter	Feature	Status	One-line note
44	See notifications in the header bell	[Implemented]	30 s polled badge, 20 newest items, own inbox only
44	Mark notifications as read	[Implemented]	one or all; no mark-unread or delete
44	Raise a notification event	[Implemented]	enqueue in caller transaction; 6-step decision order
44	Deliver messages in the background	[Implemented]	LISTEN + 2 s poll, skip-locked claim, 5 attempts; in-app only
44	Choose who is notified (switches and rules)	[Implemented]	audited; mandatory rules locked
44	Set quiet hours	[Partial]	saved but never holds in-app messages
44	Edit message wording in several languages	[Implemented]	en / ur / roman_ur; placeholder validation
44	Review the message log	[Implemented]	filters by status, channel, text
44	Patient messaging opt-outs and language	[Partial]	backend works; UI card commented out
44	Notify through a workflow transition	[Partial]	notify effect exists; no shipped transition found using it
44	Send to patients by WhatsApp, SMS or email	[Not implemented]	no patient channel; dev-log stand-in only
44	Appointment reminders	[Not implemented]	no event, no job

12 rows: 7 Implemented, 3 Partial, 2 Not implemented, 0 Unverified. The in-app bell, message rules, templates and the delivery pipeline work. Quiet hours, patient opt-outs and workflow-driven notification are Partial. Sending to patients by WhatsApp, SMS or email and appointment reminders are not built.

64.2.18 Chapter 45 — Subscriptions and onboarding

Chapter	Feature	Status	One-line note
45	Sign up and create a clinic	[Implemented]	one transaction; 14-day Professional trial
45	Verify email and resend link	[Partial]	works, but nothing is gated by verification
45	Complete the setup wizard	[Implemented]	3 steps; owner only; finish does not check the other steps
45	Trial, plans and plan limits	[Implemented]	plans in code; PLAN_LIMIT on locations and logins
45	Read-only mode after expiry	[Implemented]	402 SUBSCRIPTION_EXPIRED; 30 s cache
45	See subscription status and warnings	[Implemented]	derived status; AccountBanner
45	Record a manual payment	[Implemented]	super admin; sets plan to paid plan
45	Change plan, dates or limits	[Implemented]	super admin; audited
45	Edit the payment instructions	[Implemented]	platform setting, 2000 chars
45	Create a clinic from the Organizations screen	[Partial]	plan selector sends values the DTO rejects

10 rows: 8 Implemented, 2 Partial, 0 Not implemented, 0 Unverified. Self-signup with a 14-day trial, plan limits, read-only mode after expiry, warnings and manual payments work. Email verification is Partial because nothing is gated by it. Creating a clinic from the Organizations screen is Partial because the plan selector sends values the API rejects.

64.3 Workflow chapters

64.3.1 Chapter 46 — Clinic onboarding

Chapter	Feature	Status	One-line note
46	Self-signup and clinic creation	[Implemented]	POST /auth/register creates user, clinic, MAIN location, trial
46	Email verification	[Partial]	Informational; nothing gated by email_verified_at
46	Setup wizard	[Implemented]	finish does not check the other steps
46	Super admin creates organization	[Partial]	Endpoint works; screen form sends values the API rejects; no owner/location created
46	Create owner login for a super-admin-created clinic	[Partial]	Only through User Admin after selecting the org; sequence deduced, not run
46	Entitle features	[Implemented]	Several key features are off by default
46	Owner enables features	[Implemented]	Locked when not entitled

Chapter	Feature	Status	One-line note
46	Roles	[Implemented]	Five system roles seeded
46	Locations	[Implemented]	Edit screen sends fields the API refuses
46	Theme and navigation	[Implemented]	Nav save bumps access_version; theme save does not

10 rows: 7 Implemented, 3 Partial, 0 Not implemented, 0 Unverified. The two ways to start a clinic are both traced. The self-signup path is built. The super admin path is Partial because the form sends rejected values and no owner login or location is created with the organization.

64.3.2 Chapter 47 — Staff onboarding

Chapter	Feature	Status	One-line note
47	Employee record, assignment and role	[Implemented]	POST /employees, phone unique platform-wide
47	Staff login (employee form route)	[Implemented]	Does not check plan login limit
47	Staff login (User Admin route)	[Implemented]	Checks login limit; links all active assignments
47	Weekly schedule	[Implemented]	Overlap helper across locations never called
47	Compensation plan	[Implemented]	Commission needs billing.commissions and consulting role
47	Assign an existing employee to a second location	[Not implemented]	No endpoint; Add Employee fails on duplicate phone
47	First sign-in and access	[Implemented]	Home page from role

7 rows: 6 Implemented, 0 Partial, 1 Not implemented, 0 Unverified. Adding staff works, with one gap: an existing employee cannot be assigned to a second location, because the only creation path refuses an existing phone number. The employee-form login route does not check the plan login limit, unlike User Admin.

64.3.3 Chapter 48 — Patient journey

Chapter	Feature	Status	One-line note
48	Patient registration	[Implemented]	Phone unique per organization
48	Appointment booking	[Implemented]	Validate then insert; DB blocks overlap
48	Check-in and token	[Implemented]	Same-day, within max lateness
48	Queue and call next	[Implemented]	Smart or first-come
48	Consultation and encounter	[Implemented]	Encounter at start-consultation
48	Treatments in the workspace	[Implemented]	Doctor sees no prices
48	Prescription in the workspace	[Implemented]	Optional
48	Lab order from the consultation workspace	[Not implemented]	Created from Lab orders page instead
48	Automatic charges on completion	[Implemented]	Only with billing.invoices enabled (default off)
48	Payment and invoice issue	[Implemented]	Invoice issued at collection
48	Receipt printing	[Implemented]	Browser print layouts; no server PDF

11 rows: 10 Implemented, 0 Partial, 1 Not implemented, 0 Unverified. The patient journey from registration to receipt is built. The only break is that no lab order can be started from the consultation workspace. Automatic charges depend on the billing.invoices feature, which is off by default.

64.3.4 Chapter 49 — Dental lab case

Chapter	Feature	Status	One-line note
49	Create and submit lab order	[Implemented]	Prices frozen at submit
49	Send magic link by message	[Partial]	Provider for contact channels is a logger
49	Copy magic link	[Implemented]	Link shown once
49	Lab accepts via link or portal	[Implemented]	Creates stages and vendor bill draft
49	Production stages (lab side)	[Implemented]	Portal login required
49	Try-in rounds	[Implemented]	No appointment automation
49	Shipment receive and lost	[Implemented]	QR token
49	Approve and deliver	[Implemented]	12-month warranty
49	Patient lab invoice	[Implemented]	Separate tables and LINV numbers
49	Vendor bill review and pay	[Partial]	Header edit and manual lines API only
49	Lab-line commission	[Implemented]	Backend only; lab cost netted

11 rows: 9 Implemented, 2 Partial, 0 Not implemented, 0 Unverified. The lab case workflow is built from order to vendor payment. Sending the magic link by message is Partial because the contact channel is a log stand-in, and vendor bill header edits are backend only.

64.3.5 Chapter 50 — Daily clinic closing

Chapter	Feature	Status	One-line note
50	Queue close day	[Implemented]	Decisions required for remaining patients
50	Automatic day-end close	[Implemented]	Only empty sessions; others alert only
50	No-show marking	[Implemented]	Manual or 60 s job; job writes no event
50	Daily collection	[Implemented]	Per payment account
50	Cash close	[Implemented]	Immutable record
50	Credit expiry job 01:00	[Implemented]	Server date
50	Invariant check 02:30	[Implemented]	Log only
50	Queue cleanup 03:15	[Implemented]	
50	Inventory alerts 08:00	[Partial]	Log only

9 rows: 8 Implemented, 1 Partial, 0 Not implemented, 0 Unverified. Closing the day is built: queue close, automatic day-end close, no-show marking, daily collection, cash close and the nightly jobs. The inventory alert job is Partial because it only logs.

64.3.6 Chapter 51 — Month-end payroll

Chapter	Feature	Status	One-line note
51	Month-end automatic generation	[Implemented]	05:00 on the 1st, server time

Chapter	Feature	Status	One-line note
51	Base plus commission	[Implemented]	Negative commission carried forward
51	Settlement lock	[Implemented]	
51	Manual generation	[Implemented]	Whole location
51	Edit and delete records	[Implemented]	Paid-out commission blocks delete
51	Put on hold	[Partial]	No button
51	Mark paid	[Implemented]	No money moves
51	Payslip PDF and CSV export	[Partial]	No buttons
51	Deductions and tax	[Not implemented]	

9 rows: 6 Implemented, 2 Partial, 1 Not implemented, 0 Unverified. Month-end payroll is built from automatic generation to mark-paid. On hold, payslip PDF and CSV export have no buttons, and deductions and tax are not built.

64.3.7 Chapter 52 — Invoice correction and refund

Chapter	Feature	Status	One-line note
52	Edit preview	[Implemented]	
52	Edit and revision	[Implemented]	Optimistic revision check
52	Commission adjustment on edit	[Implemented]	Backend only by design
52	Cancel unpaid invoice	[Implemented]	
52	Unapply payment	[Partial]	Backend only; needed to cancel a paid invoice
52	Refund from advance or invoice	[Implemented]	Cannot be undone

6 rows: 5 Implemented, 1 Partial, 0 Not implemented, 0 Unverified. Invoice correction and refund work. Unapplying a payment is Partial because it has no screen.

64.3.8 Chapter 53 — Subscription expiry and recovery

Chapter	Feature	Status	One-line note
53	Trial and banners	[Implemented]	14 days
53	Grace period	[Implemented]	3 days
53	Read-only mode 402	[Implemented]	Cached 30 s per process
53	Super admin records payment	[Implemented]	Overwrites plan
53	Return to active	[Implemented]	
53	Renewal reminder email	[Not implemented]	None sent
53	Worker jobs while read-only	[Unverified]	Deduced: guard is HTTP only

7 rows: 5 Implemented, 0 Partial, 1 Not implemented, 1 Unverified. The subscription lifecycle (trial, grace, read-only, recovery) is built. Renewal reminder emails are not sent. Whether background jobs keep running while a clinic is read-only is Unverified, deduced from the guard being HTTP only.

64.3.9 Chapter 54 — Access change propagation

Chapter	Feature	Status	One-line note
54	Owner edit of role, feature or nav	[Implemented]	
54	access_version bump	[Implemented]	Subscriber plus explicit bumps
54	Redis bus to all processes	[Implemented]	15 s fallback
54	access.changed realtime event	[Implemented]	Room org:id
54	Web app refetch	[Implemented]	Focus fallback, 30 s stale
54	Theme save triggers propagation	[Not implemented]	No bump found

6 rows: 5 Implemented, 0 Partial, 1 Not implemented, 0 Unverified. Access changes propagate through the version counter, the Redis bus, the realtime event and the web refetch. Saving a theme does not trigger propagation.

64.3.10 Chapter 55 — Purchase to consumption

Chapter	Feature	Status	One-line note
55	Reach the inventory page	[Partial]	No sidebar entry
55	Create purchase order	[Unverified]	createdBy FK to employees receives a user id
55	Receive goods	[Unverified]	performedBy FK to employees receives a user id
55	Stock movement and adjustment	[Unverified]	Same FK risk
55	Patient consumption	[Not implemented]	API only, no screen
55	Low-stock alert	[Partial]	Server log only

6 rows: 0 Implemented, 2 Partial, 1 Not implemented, 3 Unverified. The purchase-to-consumption chain is the weakest workflow: the page has no sidebar entry, three steps are Unverified because of the login-id versus employee-id foreign keys, consumption has no screen and the alert only logs.

64.4 Cross-cutting chapters

64.4.1 Chapter 56 — Authentication

Chapter	Feature	Status	One-line note
56	Register a clinic (self-signup)	[Implemented]	Creates user, organization, main location and 14-day trial
56	Verify an email address	[Implemented]	Informational only; nothing is gated by it (Chapter 45 labels the same feature Partial)
56	Sign in	[Implemented]	Session of 1 or 30 days; at most 5 sessions per user
56	Choose and switch organization and location	[Implemented]	
56	Access token, refresh cookie and silent refresh	[Implemented]	Refresh secret rotates on every use; expiry fixed
56	Sign out, list and revoke sessions	[Implemented]	A revoked session's access token lives until it expires
56	Change, forgot and reset password	[Implemented]	Six-character minimum on these paths

Chapter	Feature	Status	One-line note
56	Add a password to an account without one	[Partial]	Endpoint only; no screen calls it
56	Rate limiting	[Implemented]	In memory of one API process; five endpoints
56	Security log	[Partial]	Write-only; some event types never written; no reader
56	User administration and staff logins	[Implemented]	Eight-character minimum on this path
56	Sign-in with Google or GitHub	[Not implemented]	Disabled; strategies not registered
56	The first super admin	[Partial]	Created by a manual database step

13 rows: 9 Implemented, 3 Partial, 1 Not implemented, 0 Unverified. Sign-in, sessions, refresh and rate limiting work. The gaps are the add-password screen, a write-only security log, a manual first super admin and disabled Google and GitHub sign-in.

64.4.2 Chapter 57 — Access control

Chapter	Feature	Status	One-line note
57	Five global guards	[Implemented]	JWT, TenantContext, FeatureFlag, Access, Subscription
57	Decorators and the coverage test	[Implemented]	Only a unit test enforces that every handler carries a marker
57	AccessContext, applyScope, decide and VisibleIf	[Implemented]	
57	GET /me/access	[Implemented]	
57	Frontend can(), route gate, sidebar filter and No access page	[Implemented]	
57	access_version, caches and the access.changed event	[Implemented]	5 min context cache, 15 s version cache
57	Lab portal guard	[Implemented]	Separate guard for lab members
57	Row-level security in the database	[Not implemented]	Isolation is application-level only

8 rows: 7 Implemented, 0 Partial, 1 Not implemented, 0 Unverified. The guard chain, decorators, scopes, the frontend gate and propagation are built. Row-level security in the database is not built.

64.4.3 Chapter 58 — Customization

Chapter	Feature	Status	One-line note
58	Features screen and impact preview	[Implemented]	
58	Roles builder	[Implemented]	Detail in Chapter 11
58	Per-role and clinic home page	[Implemented]	
58	Sidebar builder	[Implemented]	Clinic default layout only; per-role layouts are never written
58	Registry flags for sidebar items	[Implemented]	The ownerOrSuperAdmin flag is unused
58	Theme and branding (7 keys, per location)	[Implemented]	
58	Public login theme	[Implemented]	

Chapter	Feature	Status	One-line note
58	Settings registry beyond theme	[Partial]	Only theme keys are defined

8 rows: 7 Implemented, 1 Partial, 0 Not implemented, 0 Unverified. Features, roles, home pages, sidebar and theme customization are built. The settings registry only holds theme keys.

64.4.4 Chapter 59 — Notification mechanics

Chapter	Feature	Status	One-line note
59	Pipeline from event to in-app inbox	[Implemented]	
59	Delivery by WhatsApp, SMS and email	[Not implemented]	Dev-log stand-in provider only
59	Recipient resolvers	[Partial]	Only the lab order workflow registers one
59	Decision order and quiet hours	[Implemented]	Quiet hours never hold in-app messages
59	Message templates	[Implemented]	
59	Outbox, relay and retries	[Implemented]	Status delivered is never set

6 rows: 4 Implemented, 1 Partial, 1 Not implemented, 0 Unverified. The in-app pipeline is built; external channels are not. Only the lab workflow registers a recipient resolver.

64.4.5 Chapter 60 — Audit logging

Chapter	Feature	Status	One-line note
60	History tables (audit_logs, status_history)	[Implemented]	Trigger-protected against update and delete
60	Patients audit	[Implemented]	13 record kinds plus export and import
60	Appointments audit	[Not implemented]	
60	Employees audit	[Not implemented]	Only role definitions are audited
60	Payroll audit	[Not implemented]	
60	Organizations and locations audit	[Not implemented]	Features, theme and subscriptions are audited
60	Patient audit trail reader	[Implemented]	Needs admin.audit.view
60	Organization-wide audit viewer	[Not implemented]	
60	Retention policy	[Not implemented]	No purge or archive anywhere

9 rows: 3 Implemented, 0 Partial, 6 Not implemented, 0 Unverified. Audit coverage is the clearest cross-cutting gap. Patients are audited and a reader exists. Appointments, employees, payroll and organizations and locations are not audited, there is no organization-wide viewer and no retention policy.

64.4.6 Chapter 61 — Error handling

Chapter	Feature	Status	One-line note
61	Response envelope and global exception filter	[Implemented]	Developer message carries raw text for unexpected errors
61	Validation (whitelist, forbid non-whitelisted)	[Implemented]	
61	Error code families	[Implemented]	Two parallel version-conflict codes

Chapter	Feature	Status	One-line note
61	Idempotency keys and If-Match	[Implemented]	
61	Database constraint mapping	[Partial]	Per service; no global mapping, so some errors are generic
61	Frontend toasts, states and code maps	[Implemented]	Code maps exist only for some modules

6 rows: 5 Implemented, 1 Partial, 0 Not implemented, 0 Unverified. The envelope, validation, error codes, idempotency and frontend states are built. Database constraint errors are mapped per service, so some surface as generic errors.

64.4.7 Chapter 62 — Time, money and numbering

Chapter	Feature	Status	One-line note
62	ClockService and location timezone	[Implemented]	
62	Server-clock leftovers	[Partial]	Jobs, credit expiry, daily summary, purchase order and prescription year, leave year, lab catalog day
62	Money storage and half-up rounding	[Implemented]	Paisa integers or numeric columns
62	Exclusive tax	[Implemented]	
62	Currency label	[Implemented]	Label only; no conversion
62	document_sequences and number_sequences	[Implemented]	Safe for billing and lab documents
62	Count-plus-one and last-plus-one numbers	[Partial]	Collision risk; purchase order and prescription numbers unique across organizations

7 rows: 5 Implemented, 2 Partial, 0 Not implemented, 0 Unverified. The location clock, money rules and the two safe sequence mechanisms are built. Some jobs and older numbering still use the server clock or count-plus-one.

64.4.8 Chapter 63 — Real time and jobs

Chapter	Feature	Status	One-line note
63	Process roles api, worker and all	[Implemented]	Exactly one worker
63	Socket.IO gateway and rooms	[Implemented]	
63	Outbox relay, Redis bus, heartbeat and watchdog	[Implemented]	
63	Scheduled jobs (seven of eight)	[Implemented]	
63	Inventory alert job	[Partial]	Logs only; sends nothing

5 rows: 4 Implemented, 1 Partial, 0 Not implemented, 0 Unverified. Process roles, sockets, the relay and seven of the eight jobs are Implemented. The inventory alert job is the eighth.

64.5 Totals

The counts in this table were computed by counting the rows of the tables in this chapter with a script, not estimated.

Chapter	Subject	Implemented	Partial	Not implemented	Unverified	Total
28	Patients	19	9	2	0	30
29	Appointments	10	2	2	0	14
30	Queue	28	2	1	0	31
31	Consultation	11	2	2	0	15
32	Prescriptions	11	4	0	0	15
33	Treatment plans	7	6	1	0	14
34	Billing	31	3	0	0	34
35	Commissions	13	0	1	0	14
36	Dashboard and reports	6	3	0	0	9
37	Employees	10	3	2	0	15
38	Payroll	7	3	1	0	11
39	Inventory	0	6	1	3	10
40	Dental lab (clinic side)	26	4	0	0	30
41	Lab portal	24	3	0	0	27
42	Lab finance	17	4	0	0	21
43	Workflow engine (Config Studio)	11	1	0	0	12
44	Notifications	7	3	2	0	12
45	Subscriptions and onboarding	8	2	0	0	10
Modules (28 to 45)		**246**	**60**	**15**	**3**	**324**
46	Clinic onboarding	7	3	0	0	10
47	Staff onboarding	6	0	1	0	7
48	Patient journey	10	0	1	0	11
49	Dental lab case	9	2	0	0	11
50	Daily clinic closing	8	1	0	0	9
51	Month-end payroll	6	2	1	0	9
52	Invoice correction and refund	5	1	0	0	6
53	Subscription expiry and recovery	5	0	1	1	7
54	Access change propagation	5	0	1	0	6
55	Purchase to consumption	0	2	1	3	6
Workflows (46 to 55)		**61**	**11**	**6**	**4**	**82**
56	Authentication	9	3	1	0	13
57	Access control	7	0	1	0	8
58	Customization	7	1	0	0	8

Chapter	Subject	Implemented	Partial	Not implemented	Unverified	Total
59	Notification mechanics	4	1	1	0	6
60	Audit logging	3	0	6	0	9
61	Error handling	5	1	0	0	6
62	Time, money and numbering	5	2	0	0	7
63	Real time and jobs	4	1	0	0	5
Cross-cutting (56 to 63)		**44**	**9**	**9**	**0**	**62**
All rows		**351**	**80**	**30**	**7**	**468**

The module chapters hold 324 rows: 246 Implemented (76 percent), 60 Partial, 15 Not implemented and 3 Unverified. The workflow chapters hold 82 rows and the cross-cutting chapters 62. Adding modules and cross-cutting rows, which do not overlap, gives 386 distinct rows with 290 Implemented, 69 Partial, 24 Not implemented and 3 Unverified. The grand total of 468 includes the workflow rows, which repeat module features.

The pattern behind the numbers is consistent. The Partial label is almost always "the backend exists but no screen reaches it" or "the data is recorded but no rule reads it", so a Partial row is usually a small piece of web work or a decision, not a missing engine. The Not implemented rows are mostly features a clinic would expect from a mature product (patient messaging, reminders, attendance, deductions, audit coverage). The Unverified rows are concentrated in inventory, where a foreign-key mismatch could make writes fail at run time, and in one job-behaviour question during read-only mode. Chapter 65 lists each gap with its severity, and Chapter 67 lists the questions that decide what to build next.

Chapter 65 — Known Gaps

This chapter collects every gap found while the handbook was written: features that do not exist, features that exist in the backend but cannot be reached from the web app, settings that do nothing, data-integrity oddities and places where older documents describe something the code no longer does. It is for the owner who has to decide what to fix, build or promise before more clinics go live. Each gap states what it is, why it matters to an owner, which chapter holds the evidence, a severity and a suggested next step.

65.1 How severity is judged

Severity is judged by the effect on a live clinic, not by the effort to fix it.

- **High**: the problem can stop daily work (registering a patient, writing a prescription) or put the business at risk, and it can happen in normal use.
- **Medium**: a clinic would notice and need a workaround, or an expectation set by the product screens is not met, but daily work continues.
- **Low**: a convenience, a cleanup or a feature most clinics can live without.

No application was run while the handbook was written, so every gap was found by reading the code. Where a gap depends on behaviour that was deduced but not observed, the text says so.

65.2 Features that are not built

These features are absent. Some have a seeded feature key or a database table that suggests they exist; none of them works.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
No messages to patients by SMS, WhatsApp or email, and no appointment reminders	Clinics cannot remind patients of visits, so no-shows are handled by phone calls only. The notification pipeline already exists, but the only active channel is the in-app inbox for staff and lab members.	Chapters 44, 59	Medium	Decide on a provider (Question 3 in Chapter 67), then switch on the patient audience and add a reminder event and job.
No renewal or trial-ending messages to clinic owners	An owner learns of an expiring trial only from the banner inside the app. Expiry puts the clinic in read-only mode.	Chapters 45, 53	Medium	Send an email to the owner at fixed days before expiry.
No patient portal	Patients cannot see their own records or invoices online. The feature key was retired.	Chapters 1, 28	Low	Treat as a roadmap item; no work needed for current clinics.
No online booking and no recurring appointments	Every booking is made by staff, and a course of visits must be booked one by one.	Chapter 29	Low	Roadmap items.
No insurance claims and no online payment gateway	Insurance details can be stored (backend only) but no claim is created. Payments are recorded by staff, never collected online.	Chapters 28, 34	Low	Roadmap items; confirm whether target clinics need them.
No server-generated PDF receipts or invoices	Receipts and invoices are printed from the browser (A4 and 80 mm). They cannot be emailed or stored as files.	Chapters 34, 48	Low	Add a PDF generator only if sending receipts to patients becomes a requirement.
No attendance tracking	Payroll uses salary and commission only; there is no record of who came to work. Leave exists but is not read by payroll.	Chapter 37	Medium	Decide whether attendance is wanted at all; if so, design it with payroll deductions.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
No payroll deductions, tax or bank files	Net pay always equals gross pay, so deductions and withholding must be done outside the Platform.	Chapters 38, 51	Medium	Define the deduction rules the target market needs, then add them to payroll generation.
No deploy or continuous-integration configuration	The repository holds no Dockerfile, pipeline or hosting configuration. How the Platform is deployed lives in people's heads.	Chapter 6	High	Write down and automate the build, migration and process-role steps before more clinics depend on it.
Chairs exist as a table only	The `chairs` entity has no module, no endpoint and no screen.	Chapters 12, 26	Low	Remove the entity or build the module; the rest of the Platform does not need it.
No lab order can be started from the consultation workspace	Doctors must leave the consultation and use the Lab orders page. Lab orders carry no link to the encounter.	Chapters 31, 48	Low	Add a button that opens a lab order with the patient and doctor prefilled.
No structured allergy list and no drug-interaction check	Allergy alerts are text chips built from free-text fields. A doctor can prescribe a drug the patient is allergic to without any warning.	Chapter 31	Medium	Add a structured allergy list first; a drug check can follow.
No way to assign an existing employee to a second location	The only way to create an assignment is to create a new employee, and a phone number that already exists is refused. A doctor who works at two locations cannot be set up.	Chapters 37, 47	Medium	Add an endpoint and screen that create an assignment for an existing employee.
No SOAP note entity or note templates	Visit notes are four fixed text fields. Clinics that need structured notes cannot have them.	Chapter 31	Low	Defer until a clinic asks.
No manual commission adjustment	The source type exists in the ledger but nothing writes it, so a one-off correction to a doctor's commission needs a database change.	Chapter 35	Low	Add an adjustment screen with a reason and an audit row.
No per-location patient visit statistics	The `patient_visits` table is never written, so the endpoint always returns an empty list.	Chapters 17, 27, 28	Low	Drop the table and endpoint, or fill it from encounters.
No organization-wide audit viewer and no audit for appointments, employees, payroll, organizations or locations	The owner cannot answer "who changed this?" for bookings, staff records or payroll. Only patient records have a history screen.	Chapter 60	Medium	Extend the audit calls to the missing modules and add one viewer screen.

The most important rows are the missing deploy configuration, which is an operational risk, and the missing second-location assignment, which blocks a normal multi-branch clinic. Everything else is either a feature a clinic would ask for later or a gap that staff can work around manually.

65.3 Built but unreachable from the web app

In these cases the backend is finished, or nearly so, but no screen calls it, so the feature cannot be used by a clinic. Each one is usually a small piece of web work.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
The inventory page has no sidebar entry and nothing links to it	Stock, suppliers and purchase orders work, but a user can only reach them by typing the address. Most staff will never find them.	Chapters 39, 55	Medium	Add a sidebar entry; resolve the foreign-key question in Section 65.6 first.
Unapply a payment has no screen	An invoice that has any payment on it cannot be cancelled from the web app, because cancelling needs the payments released first. The workaround is a refund.	Chapters 34, 52	Medium	Add an Unapply action to the payment detail.
Patient treatment plans have no screen	Creating, updating and tracking plans, turning plan items into charges and the plan billing summary exist only in the backend. Doctors cannot plan a course of treatment in the Platform.	Chapters 28, 33	Medium	Build the plan screens, or remove the backend if plans are not wanted.
Payroll: put on hold, CSV export and payslip PDF have no buttons	An owner cannot give staff a payslip or export a month for an accountant from the Platform.	Chapters 38, 51	Medium	Add the three buttons; the endpoints exist.
Lab portal: charge requests and file upload have no portal control	A lab cannot ask for an extra charge or attach a file from the portal, so these happen by message or phone.	Chapters 40, 41	Medium	Add the two forms to the portal order page.
Lab portal: no notification inbox	Lab members receive in-app notifications that link into the portal, but the portal layout has no bell to show them.	Chapters 40, 41	Medium	Add the bell to the portal header.
Cancel one lab item has no screen	Stopping one item of an order in progress is not possible from the web app.	Chapter 40	Low	Add a Cancel item action on the order detail.
Recalls, referrals, feedback, insurance and the communications log	The component that shows them is never mounted, and recalls have no component at all. The data model is ready.	Chapter 28	Low	Mount the tab or drop the claim; update the older documents that call it implemented.
Blacklist a patient	The flag can only be set through the API, and it blocks nothing when set.	Chapter 28	Low	Decide what blacklisting should do (see Section 65.4), then add the control.
Override booking rules	The API flag and permission exist, but no booking screen sends it, so no one can book outside working hours or past the daily limit.	Chapter 29	Low	Add an override option for users who hold the permission.
Cancel reasons	The API stores a reason code and note; the screen sends neither and shows no confirmation.	Chapter 29	Low	Add the reason dialog.
Phone pre-check at registration	The endpoint and hook exist but no component calls them. The server rejects duplicates at save with a conflict message.	Chapter 28	Low	Call the hook on the phone field.
Credit packages and patient credits	Backend only, and credits are not a payment method.	Chapter 34	Low	Decide whether prepaid packages are wanted.
Plan billing summary, daily financial summary and several billing helpers	Endpoints and client functions exist but no component renders them. `/financial-summary` redirects to the dashboard.	Chapters 33, 34, 36	Low	Remove unused code or build the screens.
About ten dashboard widgets that still use mock data	They are not on the live dashboard but remain in the code and could be mistaken for real features.	Chapter 36	Low	Delete the mock module.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Queue: next-patient preview, queue history, the doctor's own session read and session creation	Backend only. The queue works without them.	Chapter 30	Low	Add a history screen for supervisors if audit of the queue is needed.
Prescriptions: edit one medicine line, duplicate, change history viewer, lookup by appointment	Backend only. The history read and the duplicate action are also likely to fail (see Section 65.6).	Chapter 32	Low	Fix the two defects, then add the screens.
Employee portal-access switch and some schedule calls	The endpoint exists, but the switch is set through the employee form instead. `/employees/create` exists but nothing links to it.	Chapter 37	Low	Clean up.
Lab: vendor bill header edit, manual bill lines, using lab advance on bills, impressions to the lab, service categories, specialty enablement	Endpoints exist; no buttons call them.	Chapters 40, 42, 43	Low	Build when the workflow is needed.

The Medium rows share one theme: a flow is half-finished at the boundary between backend and web. Finishing the inventory entry, the unapply action and the payroll buttons would remove most of the daily friction an owner would see.

65.4 Seeded but inert

These things look real in the catalog, a table or a setting, but nothing reads them or acts on them.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Legacy organization settings (booking defaults, advance booking days, cancellation notice, late fee percentage and old feature switches)	An owner can save these values and nothing changes. Only a location settings screen reads the old feature switches to hide sections.	Chapters 9, 14	Medium	Remove the screen or wire the rules (Chapter 67, Question 5).
Blacklisted and inactive patient flags	A blacklisted or inactive patient can still be booked, queued and billed. Staff may assume the flag protects them.	Chapter 28	Medium	Decide the intended rule and enforce it at booking and check-in.
Availability overrides and approved leave	They can be recorded and edited, but booking, validation, the calendar, the queue and payroll never read them. A doctor on approved leave can still be booked.	Chapters 29, 37	Medium	Make booking and the calendar read them.
Feature keys that gate nothing (`comms.in_app` , `comms.whatsapp` , `hr.commissions`)	Switching these on or off has no effect. Twelve more keys for unbuilt features are deactivated at every start and are hidden.	Chapters 10, 35, 44	Low	Retire the three keys or attach checks.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Notification events that are never emitted (`order.overdue` , `shipment.dispatched` , `tryin.appointment_due` , `appointment.rescheduled_lab_delay`)	Clinics can edit the wording of messages that will never be sent. There is no overdue job.	Chapters 40, 44	Low	Add emitters or remove the events.
Quiet hours, patient opt-outs and the delivered status	Quiet hours are saved but never hold in-app messages. The outbox status `delivered` is never set. The opt-out screen is commented out.	Chapters 44, 59	Low	Fix or remove.
The `queue.display.manage` permission	It is in the catalog and in the Receptionist template but nothing checks it.	Chapter 30	Low	Attach it to display management or drop it.
Email verification	The verification state is stored but nothing is gated by it.	Chapters 45, 56	Low	Decide whether unverified owners should be limited.
Dead schema and code (the table for linked external sign-ins, the deprecated refresh-token column, unused queue and database packages, per-role sidebar layouts that are never written, unused audit and history values, unused magic-link limits)	They add noise for the next developer and imply features that are not there.	Chapters 4, 13, 15, 32, 40, 58, 60	Low	Clean up in one pass.
Settings registry beyond theme	The generic cascading settings store holds only theme keys.	Chapter 58	Low	Move new settings onto it or leave as is.

The Medium rows are the ones that mislead: a screen or flag that implies protection or behaviour the Platform does not provide. Cleaning them up costs little and removes false assurance.

65.5 Data-integrity oddities

These are places where the data model can produce a wrong or failing result in normal use. Several were deduced from code and constraints and not reproduced.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Patient numbers are "count plus one", but patients can be permanently deleted	After any patient is deleted, the next registration computes a number that already exists and fails on the unique index. Because the count does not change, every later registration fails the same way. This was confirmed by reading the code.	Chapters 28, 62	High	Use the safe sequence mechanism that billing uses, and stop hard-deleting patients that have history.
Prescription numbers are unique across the whole Platform, but are numbered per organization	The first prescription of a second clinic in the same year is numbered like the first clinic's first one and violates the unique key. This was confirmed by reading the code, but not run.	Chapters 19, 32, 62	High	Make the number unique per organization, or use a Platform-wide sequence.
Purchase order numbers have the same flaw	Same cause, in inventory. A second clinic's first purchase order collides.	Chapters 23, 39, 62	Medium	Same fix.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Other count-plus-one numbers (treatment plan, legacy prescription, employee code) and unlocked "last plus one" numbering	They can repeat after a delete or when two people save at the same moment.	Chapters 28, 33, 37, 62	Medium	Move to the sequence mechanism.
Login id stored where an employee id is expected	Several writes (inventory movements and purchase orders, prescription duplication, saved doctor signatures) put a login id into columns defined as links to employees. In some tables there is no foreign key, so the value is silently wrong; where there is one, the write may fail. Not run.	Chapters 28, 32, 39, 55	Medium	Decide which id each column should hold and make the code consistent.
Deleting a patient or an employee is blocked by `ON DELETE RESTRICT` rows and fails with a generic server error	The user sees "Something went wrong" instead of a reason. A code comment claims the rows cascade, but they do not.	Chapters 16, 17, 28, 37	Medium	Catch the foreign-key error and give a clear message, or soft-delete.
Duplicate CNIC or phone number on employee edit	Reaches the database unique index and returns a generic error.	Chapter 37	Low	Check before saving.
Employee phone and CNIC are unique across the whole Platform	By design one person record can serve several clinics, but the only creation path refuses an existing phone (see Section 65.2). A second clinic cannot hire someone another clinic already registered.	Chapters 8, 37	Medium	Add the assign-existing-employee flow.
Tenant columns without foreign keys on most billing, queue and lab tables, and logical references to unknown targets (`dentist_id` , `manager_id`)	The database does not stop a row pointing at a missing or wrong organization. Nothing in the code explains the choice.	Chapters 12, 27	Low	Document the decision or add the keys.
"Append-only" tables enforced by convention only	Only three tables are protected by triggers. The billing audit log, the queue log and the ledgers can be changed by anyone with database access.	Chapters 26, 27	Low	Add triggers to the money tables.
Jobs and statistics that use the server clock	The month-end payroll month, credit expiry, daily summary, "this month" prescription counts, the lab catalog price day (fixed to one time zone) and the leave year follow the server, not the location. Wrong around midnight for locations in other zones.	Chapters 38, 62	Medium	Move jobs to the location clock or document the server zone.
The web app computes some appointment ranges in the browser time zone	The server uses the location time zone, so a user in another zone sees shifted times.	Chapter 29	Low	Use the location zone in the web app.
Location defaults of UTC and USD while organizations default to Asia/Karachi and PKR	A new branch created without explicit values gets different defaults from its clinic.	Chapters 1, 9, 14	Low	Make new locations inherit the clinic's values.
Concurrent patient imports are blocked only inside one process	With several API instances two imports can run at once.	Chapter 28	Low	Use a database lock.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Registration compares phone numbers exactly, import compares normalised numbers	The same number typed two ways can create two patients.	Chapter 28	Low	Normalise in both.
Queue policy values apply within about a minute, and per process	The screen says changes apply tomorrow. Only the smart-queue switch is fixed per day. Another instance can use old values for up to a minute.	Chapter 30	Low	Correct the screen text.
Check-in lookup and no-show job: write through GET and silent SQL	One read endpoint creates a session. The no-show job writes no queue event.	Chapter 30	Low	Log the event.

The two High rows deserve attention first because they stop core work. The patient-number problem appears after the first deletion in any clinic. The prescription-number problem appears as soon as a second clinic writes a prescription in the same year.

65.6 Unverified behaviour

These items could not be confirmed from code reading, and a real run is needed.

- Whether the foreign-key mismatches in inventory (Section 65.5) make stock movements, purchase orders and goods receipts fail at run time. Chapter 39 labels three features Unverified for this reason.
- Whether the prescription change-history read fails because it asks for a relation that is not declared, and whether duplicating a prescription fails on the doctor foreign key. Chapter 32.
- Whether background jobs keep running while a clinic is read-only. The read-only rule is an HTTP guard, so the answer is probably yes. Chapter 53.
- Whether a local PostgreSQL without SSL works with the connection settings. Chapter 7.
- Which account type to create in User Admin for a person who works only for a lab, and the exact sequence a super admin must follow to create an owner login for a clinic created from the Organizations screen. Chapters 41 and 46.
- Whether storage settings make uploaded patient documents and prescription PDFs publicly readable by address. Chapters 28 and 32.

65.7 Documentation drift

Several older documents describe a Platform that no longer exists. The outdated ones should not be used. Chapter 68 lists each conflict with evidence. The gaps that matter for planning are these.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
The root project instructions describe a frontend layout, a four-guard chain, "stub" modules and path-prefix skips that the code does not have	A new developer or an AI assistant that follows them will look in the wrong places.	Chapters 4, 5, 8	Medium	Update the instructions file to match the code.
Five historical documents (hand-off note, project overview, audit, plan and the old billing context pack) are outdated	They list fixed problems as open and built features as missing.	Chapter 68	Medium	Archive them or mark them superseded by this handbook.
The first inventory of the documents marked recall, feedback and referral as implemented through a CRM tab	The tab is never mounted. This handbook corrects that to Partial.	Chapters 28, 68	Low	Use the handbook as the source of truth.

Gap	Why it matters to an owner	Evidence	Severity	Suggested next step
Some screen texts claim behaviour that does not exist (starting a patient plan from a template, queue settings applying "from tomorrow", a lab access panel saying a link expires after a day)	Staff are told something false.	Chapters 30, 33, 40	Low	Correct the texts.
The generated data-model catalog has two known defects (shifted notes for ten inventory tables, a false trigger claim on form schema versions)	Anyone reading the catalog directly can be misled.	Chapters 23, 26	Low	Fix the generator.

Documentation drift is Medium at worst, but it compounds: every wrong document is a future wrong decision.

65.8 What to do first

The gaps fall into a clear order. First fix the two High data-integrity gaps (patient numbers after deletion and per-organization prescription numbers), because they break core work and will appear in normal use. Second, write down the deploy procedure, because it is the only High gap that is not a code defect. Third, finish the half-built flows that cost clinics time: the inventory sidebar entry, the unapply action, the payroll buttons and the lab portal forms. Fourth, remove or wire the settings that imply protection, such as the legacy organization settings and the blacklist flag. Everything else is roadmap work that depends on the answers in Chapter 67.

Chapter 66 — Risks

This chapter lists the risks an owner carries by running the Platform as it is today: risks in the data model, risks in how it is operated, observations that a regulator or a careful customer might ask about, and security risks described at category level only. It is for the product owner, who decides what to fix before more clinics rely on the Platform. Nothing in this chapter was tested by running the application; every item comes from reading the code and the generated data-model catalog.

66.1 How to read the risks

Each risk has a likelihood in normal use (Likely, Possible, Unlikely) and an impact on a live clinic (High, Medium, Low). Security items are grouped by category and deliberately contain no file paths, endpoint paths, secret values or exploit detail, because this text is shared outside the engineering team. The engineering team holds the detailed findings separately.

66.2 Data-model risks

Risk	What can happen	Likelihood	Impact	Evidence
Count-plus-one and last-plus-one numbering	After a permanent delete, or when two people save at the same moment, a patient, plan, employee or purchase order number can repeat and the save fails.	Likely for patients after any delete	High	Chapters 28, 62
Platform-wide unique keys with per-organization numbering	Prescription and purchase order numbers restart for each organization but are unique across all of them, so a second clinic collides with the first.	Likely in a multi-clinic deployment	High	Chapters 19, 32, 39, 62
Login id stored in employee-id columns	Some writes put a user id where an employee id is expected. Without a foreign key the value is silently wrong; with one the write may fail.	Possible	Mediu m	Chapters 28, 32, 39
Tenant columns without foreign keys	Most billing, queue and lab tables keep organization and location ids with no foreign key, so the database cannot reject a wrong tenant id.	Unlikely	Mediu m	Chapters 12, 27
"Append-only" promised by comments only	Only the commission ledger, the audit log and the status history are protected by triggers. The billing audit log, the queue log, the ledgers and the allocation table rely on application code.	Unlikely	Mediu m	Chapters 26, 27
Duplicate concepts	Two prescription tables, two audit mechanisms, two number-sequence mechanisms, two outbox tables and two feature-switch mechanisms coexist. A change to one can miss the other.	Possible	Mediu m	Chapter 27
Deletes blocked by RESTRICT keys	Deleting a patient or employee with history fails with a generic error.	Likely	Low	Chapters 16, 17
Schema drift between development and production	Development builds the schema from the entities; production applies a reviewed diff. Two migrations were written before later entity changes, so production needs the diff applied.	Possible	Mediu m	Chapters 6, 12
Mixed timestamp types	Older tables use timestamps without a time zone, newer ones with.	Unlikely	Low	Chapter 12

The two numbering rows are the most urgent because they fail in ordinary use. The rest are latent: they cost nothing until a clinic hits a particular edge.

66.3 Operational risks

Risk	What can happen	Likelihood	Impact	Evidence
Exactly one worker process	The worker runs the outbox relay, the notification relay and all eight scheduled jobs. Two workers would duplicate jobs; none would stop live queue updates, notifications and nightly checks. There is no supervision described.	Possible	High	Chapters 6, 63
In-memory rate limiting	Login and other limits are counted in one API process. With several instances the limit multiplies, and a restart clears it.	Likely when scaled out	Medium	Chapters 6, 56
Server time zone for jobs	No schedule sets a time zone, so jobs follow the server clock, not each location.	Likely	Medium	Chapters 6, 62, 63
No deploy or CI configuration	There is no pipeline, container file or hosting description in the repository. Builds and releases depend on individual knowledge. The built output path also differs from the one in the start script.	Likely	High	Chapter 6
Dependence on Redis	Realtime updates, access-change propagation and settings caches use Redis. Without it clinics see stale screens for up to the fallback interval.	Possible	Medium	Chapters 54, 63
Per-process caches	The subscription decision (30 seconds), access context (5 minutes), version (15 seconds) and queue policy (60 seconds) are cached per process.	Likely	Low	Chapters 30, 45, 57
Nightly jobs only log	The money invariant check and the inventory alert job write to the log. Nobody is told unless someone reads it.	Likely	Medium	Chapters 34, 39
Calendar licence	The appointment calendar uses premium resource views from a third-party library, and the licence key in use is a non-commercial one.	Likely	Medium	Chapter 29
Seeded data re-created at boot	Seeds run in the worker as well as the API and recreate a deleted system role, and re-seed starter plan templates when a clinic has none.	Possible	Low	Chapters 11, 33
Unrun test coverage of behaviour	The handbook was written by reading code. The access and billing suites exist, but the findings here were not reproduced.	Possible	Medium	Chapter 64

Of these, the single worker and the missing deploy configuration matter most, because a failure there is silent and affects every clinic at once.

66.4 Compliance-sounding observations

These are observations, not legal conclusions. A clinic or a regulator may ask about them; the Platform makes no compliance claim, and earlier screenshots that showed such claims were excluded from the handbook.

- **No retention policy.** Nothing purges or archives old sessions, security logs, audit rows, notification messages or consent signatures. Network addresses and browser strings are kept indefinitely.
- **Audit coverage is partial.** Patient records, billing, prescriptions and queue events keep history. Appointments, employees, payroll, organizations and locations do not, and there is no organization-wide viewer.
- **No row-level security.** Tenant isolation is application-level only. The queue sets database session values for tenant and location, but no database policy uses them.

- **Patient documents on external storage.** Documents, insurance images and prescription PDFs are stored with an external provider and referenced by address. Access to those addresses is not decided by the Platform's permissions.
- **Consent signatures inside the database.** Signature images are stored inline in the consent row, and immutability of consents is a convention, not a database rule.
- **Message text in logs.** The stand-in delivery provider writes message bodies and recipient addresses to the application log, which would be unsuitable once real patient channels are switched on. Outbox rows also keep rendered text.
- **No second factor or lockout.** Accounts rely on a password with a short minimum.

66.5 Security risks by category

These categories summarise findings from all chapters. They contain no locations, paths or exploit detail by design.

Category	Summary	Impact	Chapters
Tenant isolation checks missing on some endpoints	A few reads or writes do not filter by organization or location, or do not verify that referenced records belong to the caller's organization or patient scope. A few reads apply organization scope where the permission has a narrower scope.	High	19, 30, 32, 33, 34, 42
Application-level isolation only	No database row-level policies back up the application checks, so one missed check is a data leak.	Medium	8, 30, 57
Committed credentials in test scripts	Test scripts and a fixture in the repository hold a database connection and test passwords. They need rotation and removal from history.	High	Inventory of documents
Token handling and logging	A legacy realtime gateway logs token-related data and accepts any origin. Real-time connections check the token once and outlive its expiry. A revoked session's access token stays valid until it expires.	High	56, 57, 63
Input from clients trusted	The server fetches a signature image address supplied by the client. An upload route is open to any organization member. Upload size and type are only partly restricted. Some edit screens send whole rows.	Medium	28, 32, 58
Information exposure in errors	The error envelope carries raw internal messages for unexpected errors.	Medium	5, 61
Public routes by design	The waiting-room display and lab magic links need no login by design, using an identifier or a bearer token as the credential. They rely on expiry, revocation and throttling, and the throttle is in memory.	Medium	30, 40, 41
Secret reuse	The signing secret for magic links falls back to the access-token secret when its own is missing.	Medium	40, 41
Authentication hardening	Short minimum password, two different minimums on two paths, no lockout, no second factor, no logging of wrong passwords, no rate limit on two sensitive endpoints.	Medium	56, 45, 47
Authorization granularity	Some actions need only a broad read permission, and one payroll permission guards all payroll data.	Medium	37, 38, 40
Audit and retention	Missing audit for personnel and payroll data; no retention.	Medium	13, 60
Money-moving replay protection	Idempotency keys are required for most money actions but optional for one vendor payment path.	Low	42
Third-party licence	See the calendar licence in Section 66.3.	Medium	29

The high-impact rows are tenant-isolation gaps, committed credentials and token handling. The others are hardening work that lowers the chance and the blast radius of a mistake.

66.6 What to fix first

Fix the committed credentials first: rotate them and remove the files from history, because the exposure already exists. Then close the tenant-isolation gaps and the token logging, since a single missing filter lets one clinic read another's data. Third, fix the two numbering collisions, which break daily work. Fourth, document and automate deployment and decide how the single worker is supervised. After these, add audit coverage for staff and payroll and a retention policy, then harden authentication and the realtime token check. The rest can follow the roadmap.

Chapter 67 — Open Questions

This chapter lists the decisions and facts that only the product owner can supply. They come from the notes of every handbook chapter, from the questions in the generated organizations and locations chapter, from the ambiguities in the generated entity catalog and from items the code could not settle. Each question has the context and the reason the answer matters, so the answers can steer what is built, fixed or removed next.

67.1 Product direction

1. **Which patient channels will the Platform use, and who pays for them?** Context: the message pipeline is built, but only the in-app channel for staff and lab members is active. WhatsApp, SMS and email to patients are commented out, and a log stand-in takes their place. Why it matters: reminders, receipts and lab-delay messages all wait on this choice, and it also decides the privacy and logging work needed (Chapter 66).
2. **Should owners get trial-ending and expiry messages, and how many days ahead?** Context: only an in-app banner warns today; no renewal email is sent. Why it matters: expiry puts a clinic in read-only mode, so a missed warning is a lost day of work and a support call.
3. **Should the Platform have a patient portal, online booking and recurring appointments?** Context: all three were planned and their feature keys were retired. Why it matters: it decides whether the retired keys are deleted or the work is scheduled.
4. **Are insurance claims and an online payment gateway required for target clinics?** Context: insurance details can be stored (backend only) and no claim or gateway exists. Why it matters: these are large features, and the answer avoids building them early or promising them.
5. **Is attendance wanted, and should payroll have deductions, tax and bank files?** Context: payroll is salary plus commission; deductions are always zero; leave is recorded but not read. Why it matters: payroll is the most legally sensitive money flow, and clinics will ask.
6. **What should a blacklisted or inactive patient be prevented from doing?** Context: both flags are informational and block nothing. Why it matters: staff may rely on them, and the answer defines the rule to enforce at booking, check-in and billing.
7. **Should the product name be fixed?** Context: the handbook calls it "the Platform" and each clinic sets its display name; the default is "Clinic Portal". Older documents and some stored keys carry a former working name. Why it matters: it decides what is cleaned out of the code and which documents can be reused.

67.2 Clinics, locations and subscriptions

These come from the generated organizations and locations chapter.

8. **Should a clinic created by a super admin start on a trial, or stay unmanaged?** Context: unmanaged clinics have no expiry and no plan limits. Why it matters: it decides whether all clinics are subject to billing.
9. **When a clinic pays for a lower plan during a higher-plan trial, should the payment downgrade it at once?** Context: recording a payment overwrites the plan with the paid plan. Why it matters: it can silently remove features during a trial.
10. **Should deactivating a clinic also make it read-only and end sessions?** Context: today deactivation only hides the clinic at login. Why it matters: owners may still hold working sessions, and the subscription guard is not involved.
11. **Can a clinic have more than one main branch, and which becomes main when one is deactivated?** Context: the main-branch flag is not exclusive. Why it matters: reports and defaults may pick the wrong branch.

12. **Should new branches inherit the clinic's time zone and currency, and should changing the clinic's time zone update existing branches?** Context: locations default to UTC and USD while organizations default to Asia/Karachi and PKR; the setup wizard updates branches but later changes do not. Why it matters: it controls every date and amount a branch shows.
13. **Should the legacy organization settings be removed or wired?** Context: booking defaults, advance booking days, cancellation notice, late fee percentage and old feature switches are saved but read by no backend rule. Why it matters: owners may think these rules are in force.
14. **Who is the location manager meant to be, an employee or a login, and what powers should a manager have?** Context: the column's target is not stated anywhere. Why it matters: a branch-manager role needs a definite owner.
15. **Should staff without the locations permission read booking-related location settings?** Context: slot length and daily cap sit on a screen that needs the manage permission. Why it matters: receptionists may need to see the limits.
16. **Which account type should be created for a person who works only for a lab, and what is the exact sequence to create an owner login for a clinic made by a super admin?** Context: both were deduced, not run. Why it matters: onboarding instructions must be correct.

67.3 Data model and numbering

17. **Which id should the `dentist_id` columns hold (employee or login)?** Context: the teeth chart, plans, recalls and feedback store a value with no foreign key and no stated target. Why it matters: reports that join to the person need a defined meaning.
18. **Which id should doctor signatures, inventory movements and purchase orders store?** Context: code passes a login id into columns that are, or look like, employee links. Why it matters: the inventory writes may fail, and signatures are keyed per login across clinics.
19. **Should prescriptions and purchase orders be numbered per organization or across the Platform?** Context: the numbers restart per organization but are unique everywhere. Why it matters: it decides which side of the conflict to change.
20. **Should patients ever be permanently deleted?** Context: deletion breaks numbering and fails when billing exists. Why it matters: it decides between soft delete and a clear refusal.
21. **Is the missing foreign key on tenant columns deliberate?** Context: most billing, queue and lab tables omit it without comment. Why it matters: it decides whether to add keys or document the choice.
22. **Which duplicate concepts should be merged?** Context: legacy and new prescriptions, two audit mechanisms, two sequence mechanisms, per-visit tables with unclear roles. Why it matters: each duplicate doubles the work for every change.
23. **Should the append-only money tables get triggers like the commission ledger?** Context: three tables are protected; others are promised by comments. Why it matters: it is the cheapest strong guard for financial records.
24. **Should the unused tables and columns be dropped (linked sign-ins, per-visit summary, deprecated refresh token, chairs, legacy inventory table)?** Why it matters: they mislead readers and the generated catalog.

67.4 Operations and security

25. **Where and how is the Platform deployed, and who supervises the single worker?** Context: no deploy configuration exists. Why it matters: exactly one worker must run, and nothing here shows how that is ensured.

26. **Should the scheduled jobs use each location's time zone?** Context: jobs follow the server clock. Why it matters: payroll month, credit expiry and day-end alerts depend on it.
27. **What is the data retention policy for sessions, security logs, audit rows, messages and consent signatures?** Why it matters: it is the first question a regulator or clinic will ask.
28. **Is the calendar library licence suitable for a commercial product?** Why it matters: it is a legal exposure that can be solved by buying the licence or changing the component.
29. **Should audit extend to appointments, employees, payroll, organizations and locations, with one viewer?** Why it matters: owners cannot otherwise answer who changed a record.
30. **What should be done about the committed test credentials?** Why it matters: rotating them and cleaning history should be scheduled, not assumed.
31. **Should public lab links and the TV display remain unauthenticated by design?** Why it matters: it defines the acceptable exposure and the expiry rules.

67.5 Behaviour to confirm by running

32. **Do inventory writes fail because of the id mismatch?** Why it matters: it decides whether inventory can be opened to clinics at all.
33. **Do the prescription change history and duplicate actions fail?** Why it matters: they are small fixes if confirmed.
34. **Do background jobs continue while a clinic is read-only?** Why it matters: payroll and invariants could be skipped or not.
35. **Are uploaded patient documents and PDFs publicly readable by address?** Why it matters: it depends on the storage settings, not on the code.
36. **Which currency column do billing screens read, the location's or the organization's?** Why it matters: a mismatch shows the wrong currency label.
37. **Does the lab ever receive a magic-link message in practice?** Why it matters: today the contact channel is a stand-in, so links must be copied by hand.
38. **Should lab orders link to the appointment and treatment?** Context: the API accepts both, the web app never sends them. Why it matters: try-in scheduling and case reporting depend on it.
39. **Is a printable lab invoice or receipt needed?** Context: none was found. Why it matters: lab patient invoices cannot be handed to patients otherwise.

67.6 How the answers are used

The product questions (Section 67.1) set the roadmap. The clinic and numbering questions (Sections 67.2 and 67.3) decide small fixes that remove the High and Medium gaps in Chapter 65. The operations and security questions (Section 67.4) decide the order in Chapter 66. The questions in Section 67.5 need someone to run the application and watch, and they should be answered before the related work is planned.

Chapter 68 — Documentation Conflicts

This chapter records where existing documents, code comments and screen texts disagree with the code. In every case the handbook follows the code. It is for the owner and for new developers, who need to know which documents can be trusted and which should not be used. The table merges the document inventory made before the handbook was written with the conflicts that each chapter found.

68.1 Conflicts table

Document	Claim	Code reality	Evidence
Root project instructions	The frontend uses `app`, `shared` and feature `api` folders	The code uses `routes`, `components`, `lib`, `store` and feature `services`	Chapter 4
Root project instructions	There are four global guards	There are five; the subscription guard comes last	Chapters 5, 11
Root project instructions	The tenant guard skips path prefixes, and the JWT guard skips public markers	There are no prefixes; the guard uses decorators only, and the JWT guard checks a list of public routes	Chapters 5, 8
Root project instructions	Tokens carry organization context	Tokens carry user id, email, global role and session id only	Chapter 8
Root project instructions	Inventory, invoices, reports and notifications are stubs	They are implemented; only chairs is entity-only	Chapters 4, 12, 20, 23
Root project instructions	Soft deletes by an active flag in most modules	Patient delete is a hard delete	Chapter 28
Root project instructions	Some dashboard widgets use mock data	The live dashboard uses real data; the mock widgets are not imported	Chapter 36
Root project instructions	Organizations own employee role definitions	The role entity lives in employees, with logic in access	Chapter 15
Root readme	Bull queue and Google and GitHub sign-in are part of the stack	The queue has no producer or consumer, and OAuth is not registered	Chapter 4
Root readme	The architecture diagram shows four guards	Five guards	Chapter 5
Root readme	`/register` creates your account	It also creates a trial clinic, main location and ownership	Chapter 7
Root readme and backend readme	The start script runs `dist/main`	The build emits the entry file one folder deeper	Chapter 6
Environment example	Refresh, email and reset secrets are required	No code reads them; several variables the code reads are missing	Chapter 6
Access-control guide	`GET /me/catalog`	The endpoint is `GET /access/catalog`	Chapter 11
Access-control guide	Assistant template summary	The Assistant also holds the lab orders view permission at location scope	Chapter 11
Access-control test cases	FEAT-08 expects "refused"	The code returns a successful no-op	Chapter 10
Appointments and queue guide	No migration files; production must still switch from synchronize	There are three migrations, and production already runs with synchronize off plus a reviewed schema sync	Chapters 6, 12, 29, 30

Document	Claim	Code reality	Evidence
Appointments and queue guide	The worker runs one or more instances	Exactly one worker may run	Chapters 6, 30
Appointments and queue guide	Realtime fan-out uses the Socket.IO Redis adapter	It uses its own Redis publish and subscribe with a worker heartbeat	Chapter 30
Appointments and queue guide	Validate runs the same rules as create; overlap gives a coded conflict	Validate skips the past-time check and the override flag; the read-time overlap has no code, only the database path does	Chapter 29
Appointments and queue guide	API list is complete	It omits the session creation endpoint	Chapter 30
Commissions guide	Target is the smaller of allocations and net	The pre-tax part is pro-rated and capped, and a lab-cost basis reduces the base	Chapter 35
Commissions guide	One method writes commission entries	A lab commission service also writes entries for lab lines	Chapter 35
Commissions guide	Access by role	Access by the permissions `compensation.manage`, `earnings.all.view`, `earnings.own.view`	Chapter 35
Billing test cases	Truncate a coupons table; amounts assume no tax; role-based error texts	Coupons do not exist; exclusive tax applies; permission-based messages	Chapter 34
Old billing context pack	Describes pre-version-two billing	Outdated throughout	Chapter 34
Feature catalog seed	`billing.discounts` "Discounts & Coupons"; `reports.exports` as CSV and PDF	Coupons were removed; exports are Excel and PDF	Chapters 10, 36
Feature catalog seed	`comms.in_app`, `comms.whatsapp`, `hr.commissions` are active features	Nothing checks them	Chapter 10
Feature catalog seed	`inventory.alerts` gives low-stock and expiry alerts	The job only writes a log line	Chapters 10, 39
Entity catalog	Notes for ten inventory tables	Shifted one row down	Chapter 23
Entity catalog	A trigger protects published form versions	No such trigger; the service enforces it	Chapter 26
Entity catalog	`charges.source_id` is discriminated by `source_type`	The column is named `source`	Chapter 20
Entity catalog	Only the queue writes the idempotency table	Workflow, lab billing, lab orders and vendor payments also write it, and the queue clean-up deletes all rows older than a day	Chapter 18
Entity catalog	`patient_visits` is a per-location summary	Nothing writes it	Chapter 27
Entity catalog	Append-only list is complete	The allocation and journal-line tables are also called append-only without triggers	Chapter 27
Entity catalog	Migration creates the audit tables completely	It lacks the subject column and index present in the entity	Chapter 26
Migration for compensation	`charge_id` is not null	The entity allows null and adds a lab-line column and a check	Chapter 22

Document	Claim	Code reality	Evidence
Code comment on patient delete	Related rows cascade	Billing and lab rows block the delete with RESTRICT keys	Chapter 17
Code comment on employee delete	Related rows keep the row with the doctor set to null	They are RESTRICT, and the message names inventory even when an appointment blocks it	Chapter 16
Code comment on prescription numbers	Sequential per organization	Unique across the Platform	Chapter 19
Code comment on plan templates and screen text	Procedures are copied into a patient's plan	Nothing creates a patient plan from a template	Chapter 33
Code comment in billing audit entity	The repo has no general audit module	A general audit table exists	Chapter 21
Code comment on journal source	Expenses post journals	Only lab finance posts journals	Chapter 21
Code comment in payroll	Runs at 00:05 on the first	The schedule is 05:00 server time	Chapter 38
Code comments in lab modules	The portal will manage the lab catalog; the clinic activates labs until the portal ships	The portal ships and lets a lab admin act on links, but only a super admin manages the catalog	Chapter 40
Queue settings screen text	Changes apply from tomorrow	Only the smart-queue switch is fixed per day	Chapter 30
Access guard comment	A missing marker becomes a boot error	Only a unit test enforces markers	Chapter 11
First inventory document	Recall, feedback and referral are implemented through a CRM tab	The tab is never mounted; recalls have no component	Chapter 28
First inventory document	Leave and availability overrides are implemented	They can be recorded but are never read by booking or payroll	Chapter 37
First inventory document	Inventory is implemented	The module works through the API but has no sidebar entry and some writes are unverified	Chapter 39
First inventory document	Digital consent signing is unverified	It is implemented with a signature pad	Chapter 28
Outline	The booking form is a sheet; lab order comes before automatic charges	It is a dialog; lab work never creates clinic charges	Chapters 29, 48
Patients visit tab text	A visit is recorded at check-in	The encounter is created at start of consultation	Chapter 28

The table shows that most drift is in descriptions of structure (folders, guards, stubs) and in comments written before a redesign. The numbers and behaviours in the code are more stable than the words around them.

68.2 Documents and how far they can be trusted

The following documents are current or mostly current and may be used, with the corrections in the table: the generated entity catalog and the organizations and locations chapter, the access-control guide, the database migrations guide, the root readme (for the process split and counts) and the appointments and queue guide.

The following are outdated and should not be used for decisions: the hand-off note (it mentions an appointment token that no longer exists and fixed problems), the project overview (it predates access control, billing version two and the lab modules), the audit file (seven of its ten most serious findings are fixed), the plan file (its checklist is stale) and the old billing context pack (it describes replaced code). The two stock starter readmes carry no

project information. The design mock-up file in the repository root uses a former name and does not describe the current screens. The root project instructions are partly outdated and need an update before they are given to a new developer or an assistant.

68.3 The CRM tab correction

The first document inventory marked recall, feedback and referral as implemented through a CRM tab on the patient page. The code shows that backend endpoints exist but the tab component is never mounted in the patient detail page, so those features cannot be reached by a user, and recalls have no component at all. The handbook therefore labels them Partial. Any plan or sales material that repeated the earlier claim should be corrected.

68.4 Using this chapter

Treat the handbook as the source of truth and the code as the final authority. When a document in the table is edited, update the row, so it does not return as a source of confusion. Chapter 65 lists the same drift among the known gaps, and Chapter 67 asks which documents the owner wants archived.